Webhook Delivery and Retry Policy

PRD-WEB-027 | Product Requirements Document | Version 1.0 | 15 Oct 2026

Fictional validation document. All endpoints, identifiers, event names, and examples below are invented.

1. Purpose

Deliver business events to registered partner endpoints with predictable retry behavior, observable delivery state, and replay protection. This release supports HTTPS destinations only and uses HMAC signatures for request verification.

2. Functional requirements

ID	Requirement	Rule	Priority
WEB-01	Endpoint registration	Require HTTPS URL, secret reference, event types, and owner team. Reject embedded credentials in the URL.	Must
WEB-02	Signature	Generate HMAC-SHA256 over timestamp + raw request body. Include the key version in the signature header.	Must
WEB-03	Delivery timeout	Set connection timeout to 3 seconds and total response timeout to 8 seconds.	Must
WEB-04	Retry schedule	Retry eligible failures after 1, 5, 15, and 60 minutes; stop after the fifth attempt.	Must
WEB-05	Replay protection	Use event_id as idempotency identity. Redelivery must preserve the original event_id.	Must
WEB-06	Dead-letter queue	Move exhausted events to DEAD_LETTER and expose the last response summary.	Should
WEB-07	Manual replay	Operators may replay an event after checking endpoint health; replay creates a new delivery attempt, not a new event.	Should
WEB-08	Payload privacy	Do not include full payment credentials, passwords, or authentication tokens in event payloads.	Must
WEB-09	Observability	Record attempt count, HTTP status, latency, next retry time, and terminal outcome.	Should

3. Result mapping

The response code is evaluated before retry eligibility is decided. Do not treat all non-2xx responses as equivalent.

Observed result	Normalized outcome	Action
2xx	DELIVERED	Store response code and stop retrying.
408 / 429 / 5xx	RETRYABLE	Schedule the next attempt using the retry table. Respect Retry-After for 429 when present.
400 / 401 / 403 / 404	PERMANENT_FAILURE	Stop automatic retry; notify endpoint owner.
TLS validation error	CONFIGURATION_ERROR	Stop automatic retry and surface certificate guidance.
Connection timeout	UNKNOWN_DELIVERY	Retry with same event_id and require receiver idempotency.

4. Delivery-state reference

State	Entry condition	Terminal?
QUEUED	Event accepted and waiting for a worker.	No
IN_FLIGHT	An HTTP request is currently running.	No
RETRY_WAIT	Failure may be transient and another attempt is scheduled.	No
DELIVERED	Receiver returned an accepted 2xx response.	Yes
DEAD_LETTER	Attempt limit exhausted or permanent rejection occurred.	Yes
CANCELLED	An authorized operator cancelled delivery before the next attempt.	Yes

5. Operational notes

1. Validate endpoint configuration before enqueueing an event.
2. Persist the event payload and event_id before the first attempt.
3. Update delivery state with a compare-and-set operation so concurrent workers cannot schedule duplicate attempts.
4. For timeout responses, delivery may have occurred even when the sender did not receive a response.
5. Manual replay requires an operator reason and writes an audit entry.

6. Non-functional requirements

The system should process 500 events per minute at steady state. The delivery dashboard must load in under 3 seconds for 95% of requests with up to 30 days of history. Secrets must be referenced through a secret manager and never appear in logs.